

MALLAREDDY VISHWAVIDYAPEETH



FILE STORAGE PACKING OPTIMIZER..

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FILE STORAGE PACKING OPTIMIZER

Packs files into available storage blocks while minimizing wasted space and reducing the number of storage units used.

Problem Domain

Cloud Computing
Operating Systems
Data Management
Optimization

Team Members

Member 1 – Name
Member 2 – Name
Member 3 – Name
Member 4 – Name

Core Idea

Files = items
Storage blocks = bins
Goal = efficient packing

“Use storage capacity effectively while respecting file-size and placement constraints.”

THE CHALLENGE

Why storage packing needs optimization

Real-World Problem

Files have different sizes, but storage blocks have fixed capacities. Poor placement can leave unused space.

Why It Matters

Fragmentation can increase storage requirements and make resource usage less efficient.

Key Challenges

- Different file sizes
- Fixed capacity
- Placement constraints
- Large file collections
- Efficient block selection

Challenge in one line

How do we fit all files into limited storage with as little unused space as possible?



Block 1: 65% used

Block 2: 90% used

Block 3: 35% used

OUR INNOVATION

A practical heuristic-based packing engine

Core innovation: choose the right storage block for each file instead of placing files randomly.

First Fit

Put a file into the first block where it fits.

Simple and practical.

Best Fit

Choose the block that leaves the smallest remaining space after placement.

FFD + Priority Queue

Sort files from largest to smallest, then pack them. A priority queue can help select blocks efficiently.

Key features

- Less unused capacity
- Fewer unnecessary storage blocks
- Multiple packing strategies
- Efficient selection for larger storage pools

HOW IT WORKS

Input → Processing → Algorithm → Output



User journey

Enter files → check constraints → sort/prioritize → choose a block → update remaining capacity → show final packing.

THANK YOU

